

Robert McDowall ©

Gambusia affinis (Baird and Girard, 1853)

Common name: Western Mosquitofish

Taxonomy: available through ITIS

Identification: Western Mosquitofish is a small, live-bearing fish, that is dull grey or brown in color with no bars or bands on the sides, and has a rounded tail. Its body is short, its head flattened, and its mouth pointed upward for surface feeding. Distinguishing characteristics were provided by Rauchenberger (1989) and Page and Burr (1991) (although the latter authors treated the two forms as subspecies). *Gambusia affinis* and *G. holbrooki* were long considered subspecies of *G. affinis*, and were only recently recognized as separate species (Wooten et al. 1988; Rauchenberger 1989; Robins et al. 1991). Complicating matters of identification, most introductions occurred before the recent taxonomic change; furthermore, the origins of introduced stocks were usually unknown or unreported. In addition, both forms were widely available and thought to have been dispersed widely by humans. As a consequence, it often is not possible to determine if many of the earlier records represent introductions of *G. affinis* or of *G. holbrooki*.

Gambuisa affinis can be distinguished from *G. holbrooki* by the number of dorsal and anal fin elements and the shape of the gonopodium on males. Boschung and Mayden (2004) suggest the following combination of characters:

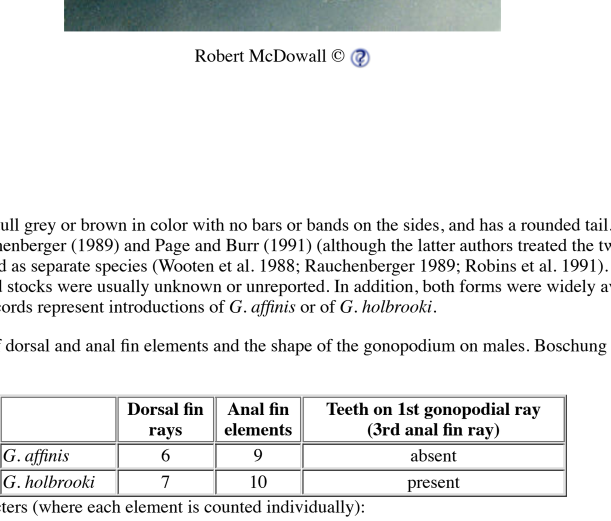
	Dorsal fin rays	Anal fin elements	Teeth on 1st gonopodial ray (3rd anal fin ray)
G. affinis	6	9	absent
G. holbrooki	7	10	present

Walters and Freeman (2000) recommend the follow combination of characters (where each element is counted individually):

	Dorsal fin elements	Anal fin elements
G. affinis	7	10
G. holbrooki	8	11

Size: 6.5 cm

Native Range: Mississippi River basin from central Indiana and Illinois south to Gulf, and Gulf Slope drainages from Mobile Bay west to Tampico, Mexico (Rauchenberger 1989; Page and Burr 1991).



Native range data for this species provided in part by NatureServe
Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Gambusia affinis* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AK	1948	1948	1	Alaska Region
AZ	1926	2023	26	Aguirre Valley; Bill Williams; Burro; Havasu-Mohave Lakes; Imperial Reservoir; Lake Mead; Lower Colorado Region; Lower Gila; Lower Lake Powell; Lower Salt; Lower San Pedro; Lower Verde; Lower Virgin; Middle Gila; Rillito; San Bernardino Valley; San Francisco; Santa Maria; Silver; Tonto; Upper Gila-San Carlos Reservoir; Upper Salt; Upper San Pedro; Upper Santa Cruz; Upper Verde; Whitewater Draw
AR	1988	2020	7	Beaver Reservoir; Elk; Little Red; Middle White; Spring; Strawberry; Upper White-Village
CA	1922	2025	67	Aliso-San Onofre; Big-Navarro-Garcia; California Region; Calleguas; Central Coastal; Coyote; Crowley Lake; Cuyama; Death Valley-Lower Amargosa; Honcut Headwaters-Lower Feather; Imperial Reservoir; Lake Tahoe; Los Angeles; Lower American; Lower Colorado; Lower Pit; Lower Sacramento; Lower San Joaquin River; Mad-Redwood; Middle Kern-Upper Tehachapi-Grapevine; Middle San Joaquin-Lower Chowchilla; Mojave; Monterey Bay; Newport Bay; Pajaro; Russian; Sacramento-Stone Corral; Salinas; Salton Sea; San Antonio; San Diego; San Francisco Bay; San Francisco Bay; San Francisco Coastal South; San Gabriel; San Jacinto; San Joaquin; San Joaquin Delta; San Luis Rey-Escondido; San Pablo Bay; Santa Ana; Santa Barbara Coastal; Santa Clara; Santa Margarita; Santa Maria; Santa Monica Bay; Santa Ynez; Seal Beach; South Fork Kern; Suisun Bay; Thomes Creek-Sacramento River; Tulare Lake Bed; Upper Amargosa; Upper Cache; Upper Coon-Upper Auburn; Upper Cosumnes; Upper Deer-Upper White; Upper Dry; Upper Pit; Upper Sacramento; Upper Stony; Upper Tule; Upper Tuolumne; Upper Yuba; Ventura; Ventura-San Gabriel Coastal; Whitewater River
CO	1967	2005	17	Big Thompson; Cache La Poudre; Colorado Headwaters; Colorado Headwaters-Plateau; Lower Yampa; Middle South Platte-Cherry Creek; Piedra; Rio Grande Headwaters; San Luis; South Platte; St. Vrain; Upper Arkansas; Upper Arkansas-John Martin Reservoir; Upper Arkansas-Lake Meredith; Upper Cimarron; Upper Colorado; Upper San Juan
CT	1935	2010	2	Outlet Connecticut River; Quinnipiac
FL	1992	1992	1	Oklawaha
GU	2004	2010	1	Guam
HI	1905	2022	6	Hawaii; Hawaii; Kauai; Maui; Molokai; Oahu
ID	1974	2007	8	C.J. Strike Reservoir; Clearwater; Lower Bear; Lower Boise; Middle Snake-Boise; Middle Snake-Succor; Upper Snake; Upper Snake-Rock
IL	1923	2017	5	Chicago; Des Plaines; Flint-Henderson; Green; Lower Fox
IN	1961	1992	4	Blue-Sinking; Lower East Fork White; Middle Ohio-Laughery; Ohio Region
IA	1987	2001	2	Flint-Henderson; Lower Iowa
KS	1945	2025	48	Arikaree; Arkansas-Dodge City; Buckner; Chikaskia; Coon-Pickerel; Cow; Crooked; Delaware; Elk; Fall; Gar-Peace; Kaw Lake; Little Arkansas; Little Osage; Lower Big Blue; Lower Cottonwood; Lower Kansas, Kansas; Lower Marais Des Cygnes; Lower Republican; Lower Smoky Hill; Lower South Fork Solomon; Lower Walnut Creek; Lower Walnut River; Marmaton; Medicine Lodge; Middle Arkansas-Lake McKinney; Middle Kansas; Middle Neosho; Middle Smoky Hill; Neosho; Neosho Headwaters; Ninnescah; North Fork Ninnescah; Pawnee; Rattlesnake; South Fork Ninnescah; South Fork Republican; Upper Cimarron-Bluff; Upper Cimarron-Liberal; Upper Cottonwood; Upper Kansas; Upper Marais Des Cygnes; Upper Neosho; Upper Saline; Upper Salt Fork Arkansas; Upper Verdigris; Upper Walnut Creek; Upper Walnut River
KY	1975	1998	10	Kentucky; Licking; Little Sandy; Little Scioto-Tygarts; Lower Kentucky; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Salt; South Fork Licking; Upper Cumberland
MA	1961	1999	2	Cape Cod; New England Region
MI	1961	2025	4	Black-Macatawa; Detroit; Great Lakes Region; Lake Erie
MN	1958	1982	4	Lower Minnesota; Twin Cities; Upper Mississippi Region; Upper Mississippi-Crow-Rum
MS	1948	1948	*	
MO	1961	2018	33	Bear-Wyaconda; Big Piney; Bourbeuse; Bull Shoals Lake; Cuivre; Current; Elk; Harry S. Truman Reservoir; Independence-Sugar; James; Lamine; Little Chariton; Lower Chariton; Lower Gasconade; Lower Missouri; Lower Missouri-Crooked; Lower Missouri-Moreau; Lower Osage; Meramec; North Fabius; North Fork White; Peruque-Piasa; Platte; Pomme De Terre; Salt; South Fabius; South Fork Salt; South Grand; Spring; The Sny; Upper Black; Upper Gasconade; Upper Grand
MT	1959	1999	8	Bullwhacker-Dog; Flathead Lake; Flint-Rock; Fort Peck Reservoir; Pend Oreille; Ruby; Upper Clark Fork; Upper Yellowstone
NE	1901	2019	31	Dismal; Harlan County Reservoir; Keg-Weeping Water; Loup; Lower Elkhorn; Lower Little Blue; Lower Niobrara; Lower North Platte; Lower Platte; Lower Platte-Shell; Lower Sappa; Lower South Platte; Medicine; Middle Niobrara; Middle North Platte-Scotts Bluff; Middle Platte; Middle Platte-Buffalo; Middle Platte-Prairie; Middle Republican; Missouri Region; North Fork Republican; Red Willow; Republican; Salt; Tarkio-Wolf; Upper Elkhorn; Upper Little Blue; Upper North Loup; Upper Republican; West Fork Big Blue; Wood
NV	1934	2023	20	Carson Desert; Central Lahontan; Havasu-Mohave Lakes; Imperial Reservoir; Lake Mead; Lake Tahoe; Las Vegas Wash; Lower Humboldt; Lower Quinn; Lower Virgin; Meadow Valley Wash; Middle Humboldt; Muddy; Ralston-Stone Cabin Valleys; Sand Spring-Tikaboo Valleys; Smoke Creek Desert; Truckee; Upper Amargosa; Walker; White
NJ	1961	2023	2	Cohansey-Maurice; Mid-Atlantic Region
NM	1950	2015	18	Carrizo Wash; Chaco; Middle San Juan; Mimbres; Pecos Headwaters; Rio Grande-Albuquerque; Rio Grande-Santa Fe; San Francisco; Taiban; Upper Gila; Upper Gila-Mangas; Upper Pecos; Upper Pecos-Black; Upper Pecos-Long Arroyo; Upper Rio Grande; Upper San Juan; Upper San Juan; Zuni
NY	1961	2023	7	Bronx; Long Island; Lower Hudson; Mohawk; Northern Long Island; Sandy Hook-Staten Island; Southern Long Island
NC	1975	1975	1	Albemarle
OH	1947	2024	18	Black-Rocky; Cedar-Portage; Licking; Little Miami; Little Muskingum-Middle Island; Little Scioto-Tygarts; Lower Great Miami, Indiana, Ohio; Lower Maumee; Lower Scioto; Muskingum; Ohio Brush-Whiteoak; Raccoon-Symmes; Sandusky; Tuscarawas; Upper Ohio-Shade; Upper Ohio-Wheeling; Upper Scioto; Walhonding
OK	1967	2025	8	Bird; Black Bear-Red Rock; Caney; Lake O' The Cherokees; Lower Neosho; Lower Salt Fork Arkansas; Lower Verdigris; Middle Verdigris
OR	1958	2012	12	Lower Malheur; Lower Rogue; Lower Willamette; Middle Willamette; Molalla-Pudding; North Umpqua; Pacific Northwest; Pacific Northwest Region; Tualatin; Umpqua; Upper Rogue; Upper Willamette
PA	2010	2025	5	Connoquenessing; Crosswicks-Neshaminy; Lower Delaware; Lower Susquehanna; Shenango
PR	1923	2007	3	Cibuco-Guajataca; Eastern Puerto Rico; Southern Puerto Rico
TN	1939	1993	1	Barren
TX	1959	1995	4	Delaware; East Galveston Bay; Lower Pecos-Red Bluff Reservoir; Pecos
UT	1931	2015	15	Great Salt Lake; Great Salt Lake; Jordan; Lower Colorado-Lake Mead; Lower Green; Lower Green-Desolation Canyon; Lower San Juan-Four Corners; Lower Weber; Southern Great Salt Lake Desert; Spanish Fork; Upper Colorado-Kane Springs; Upper Lake Powell; Upper Virgin; Utah Lake; Weber
VA	1997	2022	2	Meherrin; Middle Potomac-Catoctin
WA	1961	2020	8	Lewis; Lower Columbia-Clatskanie; Lower Snake; Lower Yakima; Middle Columbia-Lake Wallula; Pacific Northwest Region; Upper Columbia-Entiat; Upper Columbia-Priest Rapids
WV	1998	2012	3	Little Muskingum-Middle Island; Potomac; Upper Ohio-Shade
WI	1961	2013	3	Grant-Little Maquoketa; Lake Michigan; Sugar
WY	1986	1994	2	Horse; Lower Laramie

Table last updated 12/8/2025

† Populations may not be currently present.

* HUCs are not listed for states where the observation(s) cannot be approximated to a HUC (e.g. state centroids or Canadian provinces).

Ecology: Western Mosquitofish is a small live-bearing fish that feeds primarily on zooplankton and invertebrate prey at the top of the water column. Adults are known to feed on their young opportunistically (Benoit et al. 2000). This species is also well known for its high feeding capacity. Chips (2004) observed maximum consumption rates of 42–167% of their body weight per day. These organisms also require a high density of refuges to maintain populations at or near their asymptotic density (Benoit et at. 2000). Interestingly, equal numbers of male and female mosquitofish occur in the ovary and at birth while adult populations contain a disproportionately large number of females and exhibit increased male mortality after recruitment (Haynes and Cashner 1995). This is probably due to the females' ability to store sperm, a trait that renders males largely unnecessary after insemination and whose presence becomes merely increased competition for developing young.

Means of Introduction: Because of their reputation as mosquito-control agents, both *G. holbrooki* and *G. affinis* have been stocked routinely and indiscriminately in temperate and tropical areas around the world. In the United States the first known introductions of mosquitofish took place in the early 1900s (Krumholz 1948). In 1905 about 150 *G. affinis* were introduced into Hawaii from Texas to test their effectiveness in preying on mosquito larvae (Seale 1905), and by 1910 their descendants had been released into parts of Oahu, Hawaii, Maui, Kauai, and Molokai (Van Dine 1907; Stearns 1983). Also, in 1905 *Gambusia*, reportedly from North Carolina, were released into New Jersey waters for the purpose of controlling mosquitoes (Seal 1910; Krumholz 1948). In 1922 mosquitofish from Texas (900 from Austin and 300 from Hearne) were introduced into a lily pond a Sutter's Fort. That lily pond served as a hatchery used to spread *G. affinis* across California and Nevada during the 1920s and 1930s (Stockwell et al. 1996). Mosquitofish were commonly and widely introduced during the following decades by such organizations as the former U.S. Public Health Service, in large part because they were thought of as an effective and inexpensive means of combating malaria (Krumholz 1948). In more recent years, employees of many state and local health departments apparently view the use of mosquitofish to control mosquito larvae as an attractive alternative to the use of insecticides. In some areas range extensions have occurred through natural dispersal far from sites where originally introduced (e.g., Pflieger 1997).

Status: Established in most states where stocked outside its native range. Its establishment and spread in northern states is greatly restricted because the species are not, in general, cold tolerant. In most cases, overwintering in colder regions requires surfacing groundwater springs (e.g., Woodling 1985; but see Lynch 1988b). Established in Nebraska, although the populations suffer heavy (up to 99%) winter mortality (Haynes 1983). Pflieger (1997) noted that *Gambusia affinis* is more widespread and abundant in Missouri now than it was half a century ago. For instance, Pflieger indicated that, by the early 1980s, it had become established northward along the Mississippi River to Clark County, Missouri, and westward near the Missouri River to Andrew County, a range expansion attributed to a combination of natural dispersal and undocumented introductions.

Impact of Introduction: According to Courtenay and Meffe (1989), mosquitofish have had the greatest ecological impact by far of any of the introduced poeciliids. Although widely introduced as mosquito control agents, recent critical reviews of the world literature on mosquito control have not supported the view that *Gambusia* are particularly effective in reducing mosquito populations or in reducing the incidence of mosquito-borne diseases (Courtenay and Meffe 1989; Arthington and Lloyd 1989). Because of their aggressive and predatory behavior, mosquitofish may negatively affect populations of small fish through predation and competition (Myers 1967; Courtenay and Meffe 1989), and benefit mosquitos by decreasing competitive pressure from zoooplantion and predation pressure from predatory invertebrates (Blaustein and Karban 1990). In some habitats, introduced mosquitofish reportedly displaced select native fish species regarded as better or more efficient mosquito control agents (Danielsen 1968; Courtenay and Meffe 1989).

Introduced mosquitofish have been particularly destructive in the American West where they have contributed to the elimination or decline of populations of federally endangered and threatened species (Courtenay and Meffe 1989). Specific examples of their negative effects include a habitat shift and a reduction in numbers of the threatened Railroad Valley springfish *Crenichthys baileyi* in springs in Nevada (Deacon et al. 1964) and the local elimination of the endangered Sonoran topminnow *Poeciliopsis occidentalis* in Arizona (Moyle 1976a; Meffe et al. 1983, Meffe 1985). Western Mosquitofish use the same habitat as the plains topminnow *Fundulus sciadicus* and have displaced these topminnows and other species with their aggressive behavior (Whitmore 1997). The mosquitofish is also responsible for the elimination of the least chub *Iotichthys phlegethontis* in several areas of Utah (Whitmore 1997). Meffe (1983, 1985) found that mosquitofish are very aggressive, even toward larger fish. They often attack, shred fins, and sometimes kill other species. Mosquitofish are known to prey on eggs, larvae, and juveniles of various fishes, including those of largemouth bass and common carp; they are also known to prey on adults of smaller species (Meffe 1985; Courtenay and Meffe 1989). Courtenay and Meffe (1989) listed impacts on a variety of native fishes.

Introducing mosquitofish also can precipitate algal blooms when the fish eat the zooplankton grazers (Hurlbert et al. 1972), or in an increase in the number of mosquitoes if the fish eat the invertebrate predators (Hoy et al. 1972, Bence 1988). Introduced fishes, including mosquitofish, are likely at least partially responsible for the decline of the Chiricahua leopard frog *Rana chiricahuensis* in southeastern Arizona (Rosen et al. 1995). In California, *Gambusia affinis* has been documented to prey heavily on California newt *Taricha torosa* larvae (Gamradt and Kats 1996) and Pacific treefrog *Hyla regilla* tadpoles (Goodsell and Kats 1999).

Mosquitofish, and other introduced poeciliids, have been implicated in the decline of native damselflies on Oahu, Hawaii. Often the distributions of the damselflies and introduced fishes were found to be mutually exclusive, probably resulting from predation of the fish on the insects (Englund 1999).

Introductions of Western Mosquitofish have been implicated in the current restricted distribution of plains topminnow in Nebraska and may be affecting populations in Wyoming (Rahel and Thel 2004; Wyoming Fish and Game Department 2010). Schumann et al. (2015) examined the impacts of mosquitofish on populations of plains topminnow and plains killifish (*Fundulus kansae*) in Nebraska using mesocosm trials, finding increased fundulid mortailty through direct predation on larval fishes and aggression towards juveniles, as well as alteration in activity patterns and microhabitat use by the native species in the presence of mosquitofish.

Remarks: Summaries and reviews of mosquitofish introductions were provided by Krumholz (1948), Hardy (1978), Courtenay and Meffe (1989), and Dill and Cordone (1997). The identity of some mosquitofish populations introduced into selected areas is correctly known. In most cases this is because the source of the stock was reported. The Western Mosquitofish, *G. affinis*, has been documented as introduced into Arizona (Dees 1961; Miller and Lowe 1967); California (Dees 1961; Moyle 1976a; Swift et al. 1993; Dill and Cordone 1997); Florida (Chick, personal communication; Williams, personal observation); Hawaii (Stearns 1983); Illinois (Krumholz 1948; Dees 1961); Indiana (Dees 1961); Kansas (Dees 1961; Cross 1967); Massachusetts (Dees 1961); Michigan (Dees 1961); Missouri (Dees 1961; Pflieger 1997); Nebraska (Lynch 1988a, 1988b); Nevada (Miller and Acorn 1946; Dees 1961); New Jersey (Dees 1961); New Mexico (Dees 1961); New York (Dees 1961); Ohio (Dees 1961); Pennsylvania (Dees 1961); Utah (Rees 1945; Dees 1961); Washington (Dees 1961); and Wisconsin (Dees 1961).

Gambusia holbrooki was introduced into New Jersey (Fowler 1952) and into Tennessee near Knoxville and maybe to other locations as well (Starnes, personal communication). Both species have been introduced into Alabama (Boschung 1992). Shapovalov et al. (1981) indicated that both species were introduced into California, but Swift et al. (1993) argued that *G. holbrooki* never has been taken in the state and probably never was stocked. There was even mention that a hybrid between the two species was released into California waters (Dill and Cordone 1997). In their recent tome on fishes introduced into California, Dill and Cordone (1997) related their strong suspicions that pure *Gambusia holbrooki* had been introduced into that state. They based their conclusion, in part, on the importance and size of the mosquito control program in California, and the central role mosquitofish played in those attempts. However, Dill and Cordone did admit that there was no real proof that *G. holbrooki* became established in the state.

In some cases *Gambusia* stocks native to a particular region of a state were moved within the same state, in Virginia for example (Jenkins and Burkhead 1994). In contrast, Krumholz (1948) reported that mosquitofish from southern Illinois, where the species is native, were introduced into northern Illinois, an area outside its native range. Hubbs and Lagler (1958) reported that intergrades between *G. affinis* and *G. holbrooki* have been introduced into southern Michigan, but the stock did not become established. Galat and Robertson (1992) found that the Yaqui topminnow *Poeciliopsis occidentalis sonoriensis* occurring in some sites increased their fecundity in response to the presence of introduced *Gambusia*; however, the researchers noted that such habitats must also have certain environmental conditions (e.g., uniform temperatures) for maintenance of vigourous *P. o. sonoriensis* populations. Galat and Robertson concluded that conservation of some extant populations of *P. occidentalis* depends primarily on control of *Gambusia*. When compared to other *Gambusia* spp., including *G. holbrooki*, Rehage and Sih (2004) found that *G. affinis* exhibited the greatest dispersal tendency and as a result was more likely to spread to other habitats after introduction.

Introduction of the Western Mosquitofish into northern California occurred in 1922 when 600 *G. affinis* were planted in Fort Sutter lily pond. Members of this population were then introduced into the vicinities of Glenn, Kern, Coachella Valley, and Los Angeles CA during the 1920s and 1930s. In 1934, *G. affinis* were also introduced into Fallon, Nevada. From Fallon, Nevada, *G. affinis* were introduced into the following areas of Nevada: Wabuska, Garrett, Parker Ranch, and Bonham Ranch in the late 1930s and early 1940s (Stockwell et al. 1996).

Western Mosquitofish have been widely introduced outside of the continential United States for mosquito control purposes (Krumholz 1948; Purcell et al. 2012). Introductions of mosquitofish into New Zealand from the Hawaiian Islands showed a reduction in genetic diversity typical of introduced populations originating from a small number of colonizers (Purcell et al. 2012).

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Other Resources:
Gambusia Control Homepage

Distribution in Illinois - ILNHS

Gambusia affinis (Global Invasive Species Database)

Great Lakes Waterlife

US Fish and Wildlife Service Ecological Risk Screening Summary for *Gambusia affinis*

FishBase Summary

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